Dataset compression with a trained sparse basis

A real byte-level codec — transform, top-k, quantise, entropy-code, store, decode — demonstrating full-dataset recovery from ≤ 50% of the raw bytes

Setup

Each image is stored as an actual file: coefficients under a basis, top-k by magnitude, symmetric uniform b -bit quantisation, a kept-position bitmask, zlib. The decoder reconstructs from the file alone plus the basis parameter file, which is stored once per dataset and — since the 2026-07-26 accounting change — recorded per point (`basis_bytes`) but **excluded** from every per-image size below: the comparison is payload-only, mirroring the analytic baselines whose transform definitions are likewise uncounted. The codec is identical across bases; only the transform differs. Grid: $k/d \in \{0.05, ..., 0.5\}$, $b \in \{6, 8, 10\}$, all 550 images (500 train + 50 held-out test, seed-42 split). All PSNR/SSIM below are on the 50 held-out test images. The real-valued rich bases were retrained at the headline budget (1008 steps, no early stop) because the original runs saved only metrics, not parameters (see "Reproduction gate" below). The 2026-07-26 refresh added the trained DCT-IV (`dct4_ctl`, the strongest learned basis of the paper's results table) as a fourth contender; its checkpoint is stored realified — the trained sign gates sit $\sim 10^{-3}$ off the exact real point and zeroing the imaginary parts measured 0.000 dB change at every grid point (see `GATE_NOTE.md`) — and its coefficients are coded as one real component each.

Headline: recovery at half the bytes

At a total budget of 50% of raw uint8 size, the best operating points are:

dataset / basis	B/img	% raw	test PSNR (dB)	test SSIM
QuickDraw — real rich (trained)	481.8	47.1%	44.69	0.99
QuickDraw — DCT-IV (trained)	500.1	48.8%	42.28	0.97
QuickDraw — block DCT 8×8	484.6	47.3%	36.88	0.97
DIV2K-8q — real rich (trained)	31855.3	48.6%	40.67	0.98
DIV2K-8q — DCT-IV (trained)	29934	45.7%	39.31	0.98
DIV2K-8q — block DCT 8×8	31428	48%	40.96	0.98

On QuickDraw the trained real-valued rich basis recovers the dataset at 47.1% of its raw size with a 7.81 dB PSNR advantage over the identical codec built on block DCT. On DIV2K the two are at parity (−0.29 dB), consistent with the committed top-k metrics.

Key result — structural distortion at equal storage (QuickDraw). The SSIM column above understates the gap: read through the residual structural distortion $D = 1 - \mathrm{SSIM}$, the two operating points at the 50%-of-raw budget are $D_{\text{rich}} = 0.007$ versus $D_{\text{DCT}} = 0.028$, i.e.

$$\frac{D_{\text{DCT}}}{D_{\text{rich}}} \approx 4.1, \qquad \frac{\mathrm{MSE}_{\text{DCT}}}{\mathrm{MSE}_{\text{rich}}} = 10^{\Delta_{\text{PSNR}}/10} \approx 6.$$

At the **same stored bytes**, the block-DCT reconstruction loses about 4.1× more image structure — and carries 6× the squared error — than the trained real-valued rich basis. (Metric definitions in "What the advantage measures" below.)

Two independent error geometries — pixelwise ℓ^2 (PSNR) and windowed structural similarity (SSIM) — agree on a 4–6× quality gap at matched storage, which is why we mark this as the headline comparison rather than the raw SSIM scores. The scores themselves (0.99 vs 0.97) sit near the saturated top of the SSIM scale, partly because QuickDraw sketches are mostly background and any competent codec reproduces flat background well; near saturation the residual D, not the score, carries the information.

Read the ratio with the appropriate care. Both entries are means over the 50 held-out test images (per-image spread: std 0.004 for rich, 0.016 for block DCT), so $D_{\mathrm{DCT}}/D_{\mathrm{rich}}$ compares split means rather than averaging per-image ratios; the two codecs are evaluated on identical images within a single run, on the current data snapshot (see "Reproduction gate"). DIV2K is the built-in control: the identical pipeline there yields $D_{\mathrm{DCT}}/D_{\mathrm{rich}} = 0.92$ — parity — so the QuickDraw gap measures the trained basis adapting to dataset structure, not an artifact of the codec machinery.

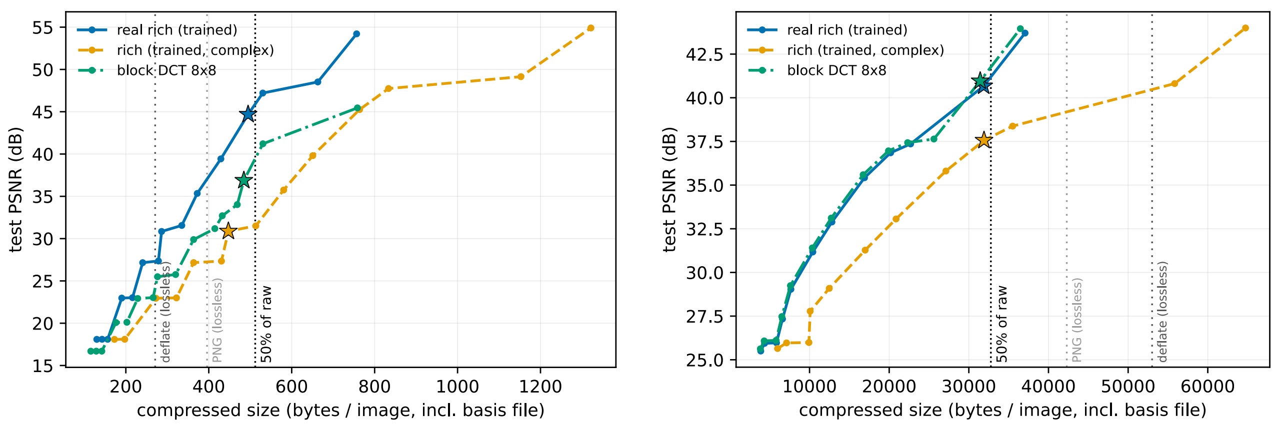

Figure 1: Rate–distortion in real bytes per image (payload only; the basis file is stored separately and excluded — note these committed panels predate the accounting change, whose per-image effect is at most the ~0.1–2% amortised basis share). Left: QuickDraw. Right: DIV2K-8q. Stars mark the best operating point within the 50%-of-raw budget (dotted vertical). Grey dotted verticals: lossless deflate / PNG references.

The paper's Fig 6 contender: trained DCT-IV

The paper's rate–distortion figure plots the trained DCT-IV against block DCT: the paper's family is full-image bases, and the DCT-IV is its strongest member on Quick Draw, so the storage section quotes it. Off its Pareto frontier in `rd_curves.json` (payload-only sizes): matched quality at 35 dB costs 383 B/img against block DCT's 474 (19.2% fewer bytes, peaking at 24% near 30 dB), and matched size at 40% of raw reads +5.42 dB (36.48 vs 31.06). At the 50%-of-raw grid budget the best DCT-IV point reaches 42.28 dB against block DCT's 36.88. Within this sweep the (blocked) real rich basis remains the strongest storage basis — its curve and the headline analysis above are unchanged, and it stays out of the paper, whose scope is full-image transforms. The DCT-IV's 23.7 KB basis file is stored separately and excluded from the sizes, like every other basis file here. On the DIV2K control the DCT-IV needs ~30% more bytes than block DCT at 35 dB and closes to within 2% at 38–40 dB — the no-gain control outcome, as with real rich.

What the advantage measures

Per image, with pixels in $[0, 1]$ (peak level $L = 1$) and the decoded image clamped to that range,

$$\mathrm{MSE}(x, \hat{x}) = \frac{1}{HW} \sum_{i,j} \left(x_{ij} - \hat{x}_{ij}\right)^2, \quad \mathrm{PSNR} = 10 \log_{10} \left(\frac{L^2}{\mathrm{MSE}} \right) = -10 \log_{10} \mathrm{MSE} \;\; [\mathrm{dB}].$$

Reported values are the means of the per-image metrics over the 50 held-out test images.

PSNR advantage. The logarithm turns error ratios into differences, so the advantage at the common byte budget B (here 50% of raw) is a multiplicative error statement:

$$\Delta_{\mathrm{PSNR}} = \mathrm{PSNR}_{\mathrm{rich}}(B) - \mathrm{PSNR}_{\mathrm{DCT}}(B) = 10 \log_{10} \left(\frac{\mathrm{MSE}_{\mathrm{DCT}}}{\mathrm{MSE}_{\mathrm{rich}}} \right).$$

QuickDraw's $\Delta_{\mathrm{PSNR}} = 7.81$ dB therefore says that at equal storage the block-DCT reconstruction carries $6\times$ the squared error ($2.46\times$ the RMS pixel error, $10^{\Delta/20}$) of the trained basis. One aggregation subtlety: since the table reports the mean of per-image decibel values, the ratio is between **geometric** means of per-image MSE across the test split, not arithmetic means. DIV2K's $\Delta_{\mathrm{PSNR}} = -0.29$ dB is a factor 0.94 — parity.

SSIM (Wang et al. 2004; scikit-image with data range 1: 7×7 windows, $C_1 = (0.01L)^2$, $C_2 = (0.03L)^2$) scores window-local agreement in luminance, contrast and structure, reaching 1 only for identical images:

$$\mathrm{SSIM}(x, \hat{x}) = \frac{1}{M} \sum_{w=1}^{M} \frac{(2\mu_x \mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)},$$

with means μ, variances σ^2 and covariance $\sigma_{x\hat{x}}$ taken over each sliding window w. Because scores sit close to 1, the informative quantity is the **residual structural distortion** $D = 1 - \mathrm{SSIM}$, and the advantage is its ratio at matched bytes:

$$\frac{D_{\mathrm{DCT}}}{D_{\mathrm{rich}}} = \frac{1 - \mathrm{SSIM}_{\mathrm{DCT}}}{1 - \mathrm{SSIM}_{\mathrm{rich}}}.$$

On QuickDraw $D_{\mathrm{DCT}}/D_{\mathrm{rich}} \approx 4.1$ — block DCT retains about $4.1\times$ the structural distortion of the trained basis at the same stored size. On DIV2K the ratio is 0.92 — parity, matching the PSNR picture.

Honest accounting

Complex coefficients cost double. The complex rich basis stores re + im per kept coefficient, so at matched bytes it operates at half the coefficient count — visible as the orange curve sitting below the real-valued blue one on both datasets (QuickDraw: 31.51 dB; DIV2K: 37.56 dB at the 50% budget). This is why the real-valued variant is the storage contender.

Lossless floor. QuickDraw sketches are mostly background: lossless deflate of the raw bytes is already down to ∼26% of raw size, and per-image PNG sits at ∼39% — both vertical references fall well left of the 50% line in the figure, comfortably below any of the trained-basis operating points shown. The QuickDraw claim is therefore basis-versus-basis at a matched lossy budget, not versus-PNG. DIV2K natural images have no such floor (deflate ∼ 81%, PNG ∼ 65% of raw) and carry the size claim directly.

DIV2K parity. On DIV2K the trained basis matches block DCT rather than beating it, consistent with the committed top-k metrics; the win case is QuickDraw.

Reproduction gate

DIV2K retraining reproduced the committed structure-tree cells exactly (≤ 0.01 dB at every keep ratio, both bases). QuickDraw did not: absolute PSNR sits 0.7–1.0 dB below the committed cells because the QuickDraw `.npy` snapshot on this machine differs from the one the committed run used — even the training-free classical baselines shift (deterministic transforms, same nominal seed-42 split), while the real_rich − block_dct_8 **gap** reproduces within 0.2 dB at every keep ratio. Details in `quickdraw_5q/checkpoints/GATE_NOTE.md`. All numbers in this section are therefore within-run comparisons on the current data snapshot.